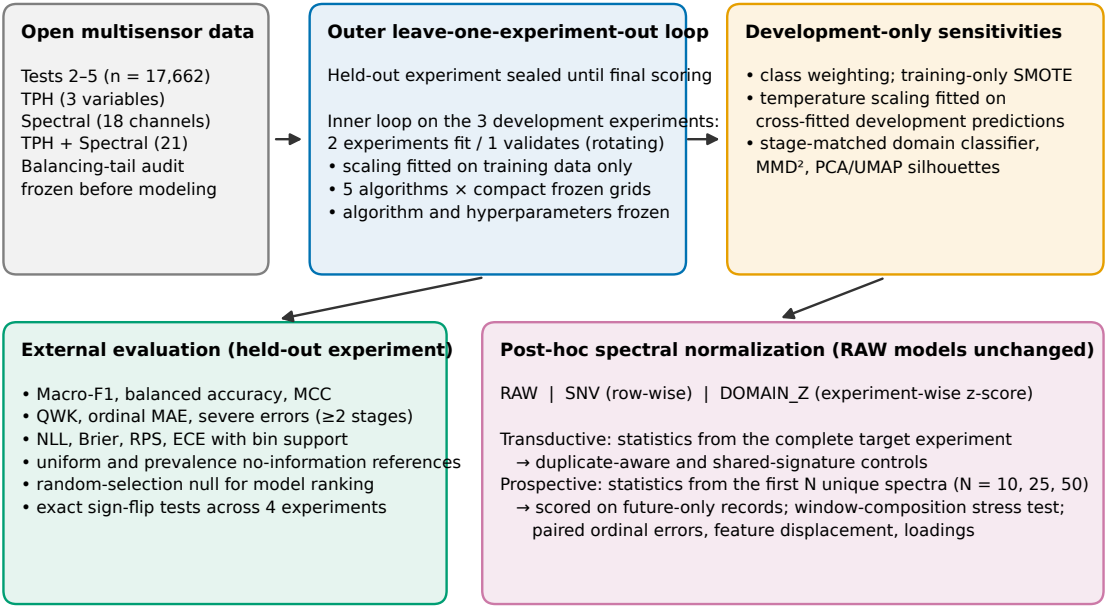
Open multisensor data
Tests 2–5 (n = 17,662)
TPH (3 variables)
Spectral (18 channels)
TPH + Spectral (21)
Balancing-tail audit
frozen before modeling
Outer leave-one-experiment-out loop
Held-out experiment sealed until final scoring
Inner loop on the 3 development experiments:
2 experiments fit / 1 validates (rotating)
• scaling fitted on training data only
• 5 algorithms × compact frozen grids
• algorithm and hyperparameters frozen
Development-only sensitivities
• class weighting; training-only SMOTE
• temperature scaling fitted on cross-fitted development predictions
• stage-matched domain classifier, MMD², PCA/UMAP silhouettes
External evaluation (held-out experiment)
• Macro-F1, balanced accuracy, MCC
• QWK, ordinal MAE, severe errors (≥2 stages)
• NLL, Brier, RPS, ECE with bin support
• uniform and prevalence no-information references
• random-selection null for model ranking
• exact sign-flip tests across 4 experiments
Post-hoc spectral normalization (RAW models unchanged)
RAW | SNV (row-wise) | DOMAIN_Z (experiment-wise z-score)
Transductive: statistics from the complete target experiment
→ duplicate-aware and shared-signature controls
Prospective: statistics from the first N unique spectra (N = 10, 25, 50)
→ scored on future-only records; window-composition stress test; paired ordinal errors, feature displacement, loadings